“White dwarfs were a type of star that astronomers had discovered through their telescopes. The mysterious thing about white dwarfs was the extremely high density of the matter inside them, a density far greater than humans had ever before encountered. Chandrasekhar had no way of knowing it when he opened Eddington’s book, but the struggle to unravel the mystery of this high density would ultimately force him and Eddington to confront the possibility that massive stars, when they die, might shrink to form black holes.

“White dwarfs are probably very abundant,” Chandrasekhar read in Eddington’s book. “Only three are definitely known, but they are all within a small distance of the Sun.... The most famous of these stars is the Companion of [the ordinary star] Sirius,” which has the name *Sirius B.* Sirius and Sirius B are the sixth and seventh nearest stars to the Earth, 8.6 light-years away, and Sirius is the brightest star in our sky. Sirius B orbits Sirius just as the Earth orbits the Sun, but Sirius B requires 50 years to complete an orbit, the Earth only one.

Eddington described how astronomers had estimated, from telescopic observations, the mass and circumference of Sirius B. The mass was that of 0.85 Sun; the circumference, 118,000 kilometers. This meant that the mean density of Sirius B was 61,000 grams per cubic centimeter—61,000 times greater density than water and just about a ton to the cubic inch. “This argument has been known for some years,” Eddington wrote. “I think it has generally been considered proper to add the conclusion ‘which is absurd.’ “ Most astronomers could not take seriously a density so much greater than ever encountered on Earth—and had they known the real truth, as revealed by more modern astronomical observations Ca mass of 1.05 Suns, a circumference of 31,000 kilometers, and a density of 4 million grams per cubic centimeter or 60 tons per cubic inch), they would have considered it even more absurd; see Figure 4.1.

Eddington went on to describe a key new observation that reinforced the “absurd” conclusion. If Sirius B were, indeed, 61,000 times denser than water, then according to Einstein’s laws of gravity, light climbing out of its intense gravitational field would be shifted to the red by 6 parts in 100,000 —a shift 30 times greater than for light emerging from the Sun, and therefore easier to measure. This redshift prediction, it seemed, had been tested and verified just before Eddington’s book went to press in 1925, by the astronomer W. S. Adams at Mount Wilson Observatory on a